# Actual monopole deficits are moduli energy in the nonlinear slow family

24 September 2026. This companion to the nonlinear slow-moduli theorem identifies its actual gauge classes and computes their deficits exactly. It connects a complete nonlinear metric family to the first-line curvature-functional theory; the Chern–Weil identity used in the connection is classical. This is part of the same slow-family result, not an independent claim of a new general curvature inequality.

## 1. Geometric setting and actual classes

Let $(B,\gamma)$ and $F$ be closed oriented surfaces of genera at least two. The base has curvature $-1$. Let $h_v$, for $v$ in a compact interval, be a smooth path of curvature $-1$ metrics on the fixed marked surface $F$, with horizontal tangent

$$k_v = \partial_v h_v, \qquad \operatorname{tr}_{h_v} k_v = 0, \quad \delta_{h_v} k_v = 0, \quad \int_F |k_v|^2_{h_v}\, dA_{h_v} = 1. \tag{1}$$

This is the positive-speed, fiber $L^2$-arclength convention of the companion, not a claim about a particular convention for a constant multiple of the Weil–Petersson metric. For a smooth function $v : B \to I$, put

$$M = B \times F, \qquad G = \gamma + h_{v(x)}, \quad A_F = \operatorname{Area}(F, h_v), \quad V = \operatorname{Vol}(M, G). \tag{2}$$

The trace condition in (1) gives a **pointwise fixed** area form $dA_F$, so $V = A_F \operatorname{Area}(B,\gamma)$.

**Proposition 1 (exact actual supply).** In the two orientations of $M$, there are fixed actual determinant classes $a_\pm$ whose harmonic representatives are

$$\omega_\pm = dA_B \pm dA_F, \qquad H_\pm = -\frac{\omega_\pm}{2\pi}, \qquad p_\pm = (a^+_\pm)^2 = a^2_\pm = A := 2\chi(M), \quad V = 2\pi^2 A. \tag{3}$$

Here $+$ on $a^+_-$ refers to self-duality in the **opposite** orientation. The forms $\omega_\pm$ are closed, nondegenerate, harmonic, and have norm squared two. They need not be parallel.

**Proof.** In the product orientation $*_G dA_B = dA_F$, so $\omega_+$ is self-dual and $\omega_-$ is anti-self-dual. Both are closed because the fiber area form is independent of $v$. They are therefore harmonic, and $\omega_-$ is self-dual after orientation reversal. The compatible almost-complex structures are $J_B \oplus J_{h_v}$ and $J_B \oplus (-J_{h_v})$. Their first Chern classes are

$$a_+ = c_1(TB) + c_1(TF), \qquad a_- = c_1(TB) - c_1(TF) = -[\omega_-]/(2\pi),$$

and $a_+ = -[\omega_+]/(2\pi)$, by the curvature $-1$ normalizations. The varying complex structures are homotopic to the constant ones along the parameter interval. In particular these are integral determinant classes, not arbitrarily chosen real classes.

In either orientation $b_+ = 1 + 2g(B)g(F) > 1$. Taubes's symplectic nonvanishing theorem supplies the actual Spin$^c$ structures. Metric-independent nonvanishing and the conformal rescaling argument give the same structure for every auxiliary positive weight, as in the polarized closure proof. Since $H_\pm$ already represents the entire class and is self-dual in its corresponding orientation, there is no unaccounted-for anti-self-dual projection. Finally $\tau(M) = 0$, and Gauss–Bonnet on the factors gives (3). □

## 2. Exact functional and Ricci-energy identities

Use tensor norms for Ricci and endomorphism norms on two-forms for Weyl. Set

$$E = \mathrm{Ric}_G + G, \quad r = \fint_M |E|^2, \quad D(v) = \int_B |dv|^2 \, dA_B, \quad \alpha = \frac{D(v)}{4V},$$

$$\mathfrak{D}_\pm = \frac{3}{2}\int_M (s^2 + 12|W^\pm|^2) - 72\pi^2 p_\pm, \qquad q_\pm = \mathfrak{D}_\pm / V. \tag{4}$$

**Theorem 2 (gauge deficit equals slow energy plus full residual).** For every metric (2), without a small-amplitude hypothesis,

$$\boxed{q_+ = q_- = 9\alpha + \frac{9}{2}r = \frac{9}{4V}D(v) + \frac{9}{2}\fint_M |\,\mathrm{Ric}_G + G|^2.} \tag{5}$$

The relative Einstein defect is exactly

$$\boxed{\mathscr{E}(G) = \frac{4r - \alpha^2}{4 + 2\alpha + r}, \qquad r \geqslant \alpha^2.} \tag{6}$$

Thus both $q_\pm = 0$ and $\mathscr{E}(G) = 0$ force $v$ constant; the resulting metric is a hyperbolic product. The assertion concerns this full block family, not all metrics on $B \times F$. The derivative budget of the actual harmonic forms is also exact:

$$\int_M |\nabla \omega_{\ \pm}|^2 = \tfrac{1}{2}D(v), \qquad q_\pm = \tfrac{9}{2}\left(\fint_M |\nabla \omega_{\ \pm}|^2 + r\right). \tag{5a}$$

**Proof.** The exact Ricci calculation of the companion gives

$$s_G + 4 = -\frac{1}{4}|k_v|^2 |dv|^2, \qquad E_{HH} = -\frac{1}{4}|k_v|^2 dv \otimes dv. \tag{7}$$

Unit fiber speed implies $\fint s = -4 - \alpha$, and direct expansion gives

$$R := \int_M |\,\mathrm{Ric}\,|^2 = V(4 + 2\alpha + r). \tag{8}$$

LeBrun's Chern–Weil identity, separately in either orientation, is

$$\mathfrak{D}_\pm = \frac{9}{2}\big[R - 8\pi^2(2p_\pm - T_\pm)\big]. \tag{9}$$

Here $T_\pm = 2\chi \pm 3\tau = A = p_\pm$ and $8\pi^2 A = 4V$. Equations (8)–(9) prove (5).

Pointwise $E_{HH}$ has rank at most one and its trace equals the entire trace of $E$; the vertical residual is trace-free. Hence $r \geqslant \fint (\mathrm{tr}\, E)^2 \geqslant \alpha^2$. Direct minimization over nonzero Einstein constants gives $\mathscr{E} = 4 - (\int s)^2/(RV)$. Inserting (8) proves (6). Its numerator is at least $3r$. Thus zero relative defect forces $r = \alpha = 0$, and (1) forces $dv = 0$. This also rules out a different nonzero Einstein normalization inside this family; its scalar curvature is negative, so Ricci-flatness is not an omitted case.

For (5a), use orthonormal base vectors $X_i$, fiber vectors $e_a$, and $S_i = \frac{1}{2}h^{-1}k_v\,\partial_i v$. The covariant derivatives of $\omega_{\ \pm}$ in horizontal directions vanish. In a vertical direction their only nonzero components are mixed base–fiber ones. Squaring these gives $2\sum_i |S_i|^2$: the cross contraction vanishes since the $S_i$ are proportional to one symmetric trace-free matrix, which is orthogonal to its product with the fiber complex structure. Hence $|\nabla \omega_{\ \pm}|^2 = \frac{1}{2}|k_v|^2|dv|^2$. Integrating the unit-speed condition proves (5a). □

## 3. The precise missing global quantity is a scalar spectral scale

Let $\lambda_1 = \lambda_1(B,\gamma) > 0$ and $\mathrm{Var}_B(v) = \fint_B (v-\bar{v})^2$. The companion's full nonlinear energy identity implies

$$r \geqslant \frac{1}{4V}\int_B (\Delta_\gamma v)^2, \qquad \Delta_\gamma = -\operatorname{div}\nabla. \tag{10}$$

Consequently

$$\boxed{q_\pm \geqslant \frac{9\lambda_1}{4A_F}\,\mathrm{Var}_B(v), \qquad r \geqslant \frac{\lambda_1^2}{4A_F}\,\mathrm{Var}_B(v).} \tag{11}$$

Both powers of $\lambda_1$ are attained in order by the actual cyclic family below. These bounds do not replace $\lambda_1$ by a genus-independent constant.

There is also an input-to-saturation implication in this family:

$$\boxed{q_\pm \leqslant \left(\frac{9}{2} + \frac{9}{\lambda_1}\right) r.} \tag{12}$$

Indeed $\int (\Delta v)^2 \geqslant \lambda_1 D(v)$, so (10) gives $\alpha \leqslant r/\lambda_1$; now use (5). If only the scale-free relative defect is supplied, then

$$\mathscr{E} < \frac{3}{7} \quad\Longrightarrow\quad r \leqslant \frac{7}{3}\mathscr{E},$$
$$q_\pm \leqslant \left(\frac{21}{2} + \frac{21}{\lambda_1}\right)\mathscr{E}, \qquad \mathrm{Var}_B(v) \leqslant \frac{28A_F}{3\lambda_1^2}\mathscr{E}. \tag{13}$$

To check the implication, (6) and $\alpha \leqslant \sqrt{r}$ give $\mathscr{E} \geqslant 3r/(4 + 2\sqrt{r} + r)$. This increasing function equals $3/7$ at $r = 1$; for $r < 1$, it is at least $3r/7$.

Equations (11)–(13) distinguish curvature-functional saturation from constancy of the actual global moduli parameter. They eliminate the auxiliary-form supply assumption and the separate near-saturation assumption here, using the genuine geometric family and its scalar spectrum. They make no analogous claim for unrestricted near-Einstein metrics without a surface fibration or without (1).

## 4. Variable base curvature and the complete cscK equality family

The same construction also allows an arbitrary base metric $\gamma$ on the fixed surface of genus at least two. Define

$$\lambda_B = -\fint_B K_\gamma = \frac{4\pi(g(B)-1)}{\mathrm{Area}(B,\gamma)} > 0.$$

The actual classes remain $c_1(TB) \pm c_1(TF)$, but their entire harmonic representatives are now $-(\lambda_B dA_B \pm dA_F)/(2\pi)$. Only their self-dual parts in the corresponding orientations enter the functional:

$$H^+_\pm = -\frac{\lambda_B + 1}{4\pi}\omega_{\pm}, \quad p_\pm = \frac{(\lambda_B+1)^2 V}{8\pi^2}, \quad T_\pm = A = \frac{\lambda_B V}{2\pi^2}. \tag{17}$$

In particular a nonzero negative projection is retained when the two factor scales differ; it is not dropped as though every actual class were self-dual.

Put $X = |dv|^2$, and let $\Xi(v) \geqslant 1/(16A_F)$ be the full normal-acceleration coefficient in the companion. Direct expansion of all its Ricci blocks and (9) give

$$\boxed{\begin{aligned} q_+ = q_- = \frac{9}{2V}\Big[2A_F \int_B (K_\gamma + \lambda_B)^2 - \frac{1}{2}\int_B K_\gamma X \\ + \frac{1}{4}\int_B (\Delta v)^2 + \int_B \Xi(v) X^2\Big]. \end{aligned}} \tag{18}$$

For verification before completing the square, the Ricci energy is

$$R = 2V + 2A_F \int_B K_\gamma^2 - \tfrac{1}{2}\int_B K_\gamma X + \tfrac{1}{4}\int_B (\Delta v)^2 + \int_B \Xi(v) X^2,$$

and $8\pi^2(2p_\pm - T_\pm) = 2(1+\lambda_B^2)V$. These prove (18) without invoking an unproved geometric limit.

If $K_\gamma \leqslant -\kappa_0 < 0$, every displayed term in (18) is nonnegative and

$$q_\pm \geqslant \frac{9A_F}{V}\int_B (K_\gamma + \lambda_B)^2 + \frac{9\kappa_0}{4V}\int_B |dv|^2. \tag{19}$$

Equality $q_\pm = 0$ is equivalent to constant $v$ and constant base curvature. The resulting product is cscK, and is Einstein only if $\lambda_B = 1$. Thus the complete classical equality family is preserved, rather than incorrectly replacing it by the KE family.

For the actual pointwise Yau26 input $\delta_-(G) < 1$, the horizontal Ricci eigenvalue perpendicular to $dv$ is exactly $K_\gamma$. Consequently $|K_\gamma + 1| \leqslant \delta_-$, so (19) applies with $\kappa_0 = 1 - \delta_-$. This also makes $1 - \delta_- \leqslant \lambda_B \leqslant 1 + \delta_-$. Equations (9),(17) and $R/V \leqslant 4(1+\delta_-)^2$ yield the concrete upper bound

$$q_\pm \leqslant 54\delta_- + 9\delta_-^2. \tag{20}$$

Thus functional saturation is automatic from pointwise Ricci pinching on this actual block family, even with variable base curvature. What is not automatic is global constancy of its moduli parameter: the scalar spectral scale in (11)–(13) still matters.

## 5. Fixed moduli oscillation with exact periods and vanishing defects

Fix a hyperbolic base $B_0$ and a nonzero primitive integral harmonic one-form $\eta$. Let $B_m$ be its connected degree-$m$ cyclic cover. On the infinite cyclic cover, choose $z$ with $dz = \eta$, $z(\tau x) = z(x) + 1$. Take a fixed $a > 0$ inside the interval (1), after translating its center, and set

$$v_m = a\cos(2\pi z/m). \tag{21}$$

For $m \geqslant 3$, summing over the deck translates gives exactly

$$\bar{v}_m = 0, \qquad \mathrm{Var}_{B_m}(v_m) = a^2/2,$$

$$D(v_m) = \frac{2\pi^2 a^2}{m}\int_{B_0} |\eta|^2, \quad \int_{B_m} (\Delta v_m)^2 = \frac{8\pi^4 a^2}{m^3}\int_{B_0} |\eta|^4. \tag{22}$$

Write $V_0 = A_F\,\mathrm{Area}(B_0)$. The complete block formula, with the fixed smooth path on a compact interval, gives

$$\delta_-(G_m) = O(m^{-2}), \quad r_m = \Theta(m^{-4}), \quad \mathscr{E}(G_m) = \Theta(m^{-4}),$$

$$\boxed{q_{+,m} = q_{-,m} = \frac{9\pi^2 a^2}{2V_0 m^2} \int_{B_0} |\eta|^2 + O(m^{-4}).} \tag{23}$$

The lower bound for $r_m$ is (10) and (22); its upper bound follows from $|dv_m| + |\nabla^2 v_m| = O(m^{-1})$ together with the sharper $\Delta v_m = O(m^{-2})$ and the exact absence of a mixed block. Equivalently the full nonlinear identity has only $(\Delta v_m)^2$ and bounded multiples of $|dv_m|^4$. The two-sided estimate for $\mathscr{E}$ follows from (6). The cyclic-cell estimate in I33 and the Rayleigh quotient in (22) give $\lambda_1(B_m) = \Theta(m^{-2})$, confirming the orders in (11). There is no claim that $v_m$ itself is an eigenfunction: $|\eta|^2$ need not be constant.

In particular all of the following hold simultaneously:

- the two gauge classes are actual and have no negative-square cost;
- their harmonic representatives are the **exact** nonvanishing constant-length forms (3), with no period error;
- pointwise Ricci defect and both normalized functional deficits vanish;
- even the unnormalized Ricci-error energy is $O(m^{-3})$;
- the marked moduli oscillation in (22) stays fixed and nonzero.

This is a concrete sharpness statement for upgrading average gauge structure to global constancy of the supplied fibration. It is not a counterexample to Yau26: every $B_m \times F$ already carries exact hyperbolic product metrics. The companion proves separation from fiber-preserving product outputs; neither note asserts separation from **all** Einstein metrics and arbitrary diffeomorphisms. Exact periods, absence of zeros, and small local derivative energy do not alone remove the long scalar slow modes.

## 6. Sources and what is additional

- C. H. Taubes, *The Seiberg–Witten invariants and symplectic forms*, Mathematical Research Letters 1 (1994), 809–822, main theorem, supplies actual classes in (3). The new argument does not claim to invent canonical-class nonvanishing.
- C. LeBrun, *Ricci curvature, minimal volumes, and Seiberg–Witten theory*, Propositions 4.1–4.2 and equation (17), supply the classical identity (9) and its gauge interpretation.
- The new nonlinear slow-family calculation supplies the full Ricci blocks and (10), retaining the normal acceleration terms rather than dropping them after projection.
- The previous cyclic inverse theorem supplies the order of the scalar spectral scale. Here the amplitude is fixed, so the new family is not confined to its shrinking $O(m^{-2})$ repair neighbourhood.

The additional connection is the exact formula (5), with actual classes and a real, fully nonlinear metric family, and its sharp spectral consequences (11)–(23). No priority or optimality beyond the stated orders on this family is claimed.